7 **(a)** State what is meant by an electric field.

.....

..... [1]

(b) A plastic rod is rubbed with a cloth. The plastic rod becomes negatively charged and the cloth becomes positively charged.

(i) Explain why.

.....

.....

..... [2]

(ii) The negatively charged plastic rod is suspended by an insulating thread. Another negatively charged plastic rod is brought close to the suspended rod.

State what happens to the suspended plastic rod.

..... [2]

(c) **(i)** Define the kilowatt-hour (kW h) in words.

.....

..... [1]

(ii) A small lamp illuminates an electric oven. The lamp has an output power of 25 W and operates for 220 hours in one year. The p.d. across the lamp is 230 V.

1. Calculate the energy transferred by the lamp in one year. Give your answer in kW h.

energy = ... kW h [2]

2. Calculate the current in the lamp.

current = ... [2]

[Total: 10]

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